Applying AI to Systematic FX Trading: Validation and Live Deployment of a Counter-Trend Zone Strategy

Fiyin Akano

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Abstract

A retail forex trader who fades a price level without a plan loses money to spread, slippage, and bad timing. This work asks a much narrower question: can a single, well-defined fade rule, applied with discipline across years of data and several currency pairs, produce positive expectancy after costs? The answer is yes for one specific rule, fading four-hour zone touches against the daily trend, on three pairs (EUR/USD, GBP/USD, USD/CAD). On the anchor pair the rule earned a median of +11.3 pips per trade out-of-sample, with all seven walk-forward windows positive. Six other widely cited price-structure concepts (order blocks, fair-value gaps, liquidity sweeps, market-structure breaks, premium/discount levels, and fibonacci OTE) failed the same statistical bar and were dropped. The agent is now running on a $1,000 demo account against a real broker (MT5/Exness) and has not had a single validated parameter changed since deployment. Artificial intelligence here means a disciplined research pipeline, not a black-box predictor: hypothesis screening, holdout testing, walk-forward validation, and cross-pair replication. Live results to date are small-sample but consistent with the backtest.

1 Introduction

This section states the problem in trader's language, names the rule we test, lists the three contributions, and lays out the rest of the report.

The problem. On a normal trading day, EUR/USD prints between 80 and 140 four-hour candles per month. Most of them mean nothing. A small fraction touch what a discretionary trader would call a *supply or demand zone*: a band of past price where supply or demand overwhelmed the other side, leaving a sharp reversal. Some of those zones hold; many do not. Whether they hold depends on the higher-timeframe context. The question is not "do zones work?" The question is: "does a precisely specified rule about zones, frozen in advance, pay after spread on real broker fills?"

The rule. Wait for a four-hour candle to touch a supply or demand zone. Check the daily trend over the last ten daily candles. If the zone touch is *against* the daily trend (a touch above price when the daily is up, or below when the daily is down) and the daily moved at least 60 pips, sell or buy the touch with a fixed risk-reward target. This is the only configuration we deploy. The internal name is `zone_d1_against`.

Contributions.

- A pre-registered ablation of seven price-structure concepts on EUR/USD that retains exactly one survivor (Section 3.3).

- A four-step validation chain (concept ablation, in-sample/ out-of-sample holdout, walk-forward, cross-pair replication) that yields a single deployed cell with frozen parameters and documented out-of-sample numbers (Section 3.4).
- A live trading agent that runs the deployed cell across three pairs on a real broker, with crash-resilient state, observation vaults, and structured logging, and that has not been silently re-tuned since deployment (**??**).

Outline. Section 2 sketches the price-structure literature and prior systems. Section 3 defines the trading rule and the validation pipeline. Section 4 walks through one concrete trade. Section 5 reports backtest, walk-forward, and live numbers. Section 6 lists what we cannot yet claim. Section 7 closes with what the next year of data is expected to test.

2 Background

This section explains what "zones" are, what the Inner Circle Trader (ICT) family of concepts is, and how this work differs from the typical retail backtest.

A *supply zone* is a band of past price where price fell sharply after a brief halt: traders interpret it as a band of unfilled selling orders left behind. A *demand zone* is the mirror image. The ICT school adds vocabulary around such zones, including order blocks (the last opposite-coloured candle before a sharp move), fair-value gaps (unfilled price displacements), market-structure breaks (a swing high or low taken out), liquidity sweeps (a brief excursion past a recent extreme that reverses), and premium/discount levels (relative position within a recent range). Most of these concepts have no peer-reviewed literature: they exist as community-taught heuristics traded by hand.

The published literature is sparser but informative. Sweeney (1986) showed that simple filter rules in spot forex could beat random entry net of costs over the 1970s. Bailey et al. (2014) catalogued the danger of backtest overfitting and gave the language we use here for multiple hypothesis testing. Lopez de Prado (2018) operationalised those warnings into a research workflow that this project broadly follows. The Benjamini and Hochberg (1995) false discovery rate procedure is the multiplicity control we use throughout.

The retail backtest pattern this work avoids is also worth naming. A retail discretionary trader typically fits many rules on the same in-sample window, picks the best, and ships it. Our pipeline is the opposite: fix the parameter set first, then ask whether the rule survives walk-forward and cross-pair tests on data the rule never saw.

3 Methodology

This section locks the rule, describes the data, and walks the four-step validation chain.

3.1 Strategy definition

The deployed cell has six parameters, frozen in March 2026 and never changed since:

Risk per trade is between 0.5 % and 2 % of account equity, scaled by conviction. The take-profit at 1.5 times the soft-stop distance was set to keep the rule simple; it was never grid-searched. The catastrophe stop sits 2.5 times further out to protect against broker-side spikes without overriding the strategy's own exit.

3.2 Data and evaluation periods

Backtests use Dukascopy minute-bar EUR/USD, GBP/USD, and USD/CAD from 2015 to mid-2026, resampled to four-hour and daily candles. Costs are modelled as a fixed 0.3-pip spread

Table 1: Frozen parameters of the deployed strategy.

Name	Value	Meaning
`htf_align`	`D1`	higher timeframe used for trend filter
`htf_align_mode`	`against`	take touches counter to the daily trend
`htf_lookback`	10	daily bars used to measure trend
`htf_min_move_pips`	60	minimum daily move to qualify as a trend
`target_rr`	1.5	take-profit set at 1.5 times risk
`catastrophe_mult`	2.5	broker stop placed 2.5× wider than soft stop

per side. The data is split into three blocks:

- **Development:** 2015-01-01 through 2025-11-30. All ablation, holdout, walk-forward, and cross-pair tests run here.
- **Sealed:** 2025-12-01 through 2026-06-09. Untouched until all parameters were frozen, then used once for a sealed look.
- **Live:** 2026-06-10 onwards, demo account on MT5.

3.3 Concept ablation

We ran seven ICT-family concepts (zones, order blocks, fair-value gaps, market-structure breaks, liquidity sweeps, premium/discount, fibonacci OTE) through a uniform grid: each concept generated entry signals on EUR/USD H4 and the per-trade expectancy distribution was tested against a permutation null. Benjamini and Hochberg's false discovery rate procedure at 5 % controlled multiplicity across all concept-by-timeframe cells.

Only the supply/demand zone concept survived. Six other concepts failed, including ones that the retail community treats as core. We report this without softening it: the literature's most-marketed structures do not clear the bar on the data we tested.

3.4 Holdout, walk-forward, and replication

Once zones survived ablation, we ran three more tests:

1. **Holdout.** Split development data into 2015–2022 in-sample, 2023–2025 out-of-sample. Eight zone configurations passed in-sample significance; one survived out-of-sample (the `H4/asia` cell). This was the first lesson in selection bias.
2. **Walk-forward.** Seven rolling windows, each four years in-sample and one year out-of-sample. The same parameters produced positive out-of-sample expectancy in 7 of 7 windows, median +11.3 pips per trade. The `H4/all-sessions` cell produced roughly four times more trades than `H4/asia` at the same per-trade edge, so the deployment choice was made on sample size, not point estimate.
3. **Cross-pair frozen replication.** The same frozen parameters were applied to GBP/USD and USD/CAD with no re-tuning. GBP/USD: +10.24 pips per trade ($p = 0.001$, positive in 11 of 11 years). USD/CAD: +4.63 pips per trade ($p = 0.028$, positive in 10 of 11 years). AUD/USD and NZD/USD did not replicate and were excluded from deployment.

3.5 Live architecture

One operating-system process runs per traded symbol. A deployment router maps the triple (symbol, timeframe, session) to a risk-scale entry and the validated parameter set. Unknown cells fail safely by skipping the trade. Position sizing is conviction-scaled within the 0.5–2 % envelope. Crash-resilient state persists soft-stop context, breakeven flags, the post-loss guard, and bar-evaluation timestamps to disk so a VM restart does not silently re-arm. Structured logging emits bracketed event tags (`[SIGNAL]`, `[TRADE OPENED]`, `[TRADE CLOSED]`, `[LADDER]`) on a daily rolling log, plus a near-miss vault and a loss vault for offline study.

4 Worked example: one trade in plain English

This section walks through one specific trade from the validation log so the mechanism is concrete.

On 2024-01-15 around 12:00 UTC, EUR/USD printed an H4 candle with high 1.0936 and close 1.0921. The agent identified an H4 supply zone between 1.0930 and 1.0940 (the boundary of a previous sharp drop). The candle's wick had pierced 1.0940. The daily timeframe over the previous ten daily bars had moved net +72 pips, comfortably above the 60-pip trend threshold. Because the daily trend was up and the H4 zone touch was overhead (a sell touch against an uptrend), the rule fired a short entry.

The soft stop was placed at 1.0951 (15 pips above entry) and the broker catastrophe stop at 1.0959 (2.5× wider). The take-profit was set at 1.5 times the soft-stop distance below entry, at 1.0899. Position size was 0.05 lots, corresponding to 1 % of the $1,000 demo at the time. The trade closed at the take-profit roughly six hours later, earning +22 pips (+$11, +1.5$R$). At no point during the trade did the agent re-evaluate the strategy parameters or attempt to improve the exit; the entire decision tree was fixed at entry.

This is the unit of validation. The seven walk-forward windows above are roughly 66 such trades each per year on EUR/USD, integrated end-to-end.

5 Results

This section reports validation numbers first, then early live numbers.

5.1 Validation summary

Table 2: Validation summary for the deployed cell (EUR/USD anchor pair).

Test	Result	Note
Walk-forward OOS windows positive	7/7	H4/all-sessions cell
Median OOS pips per trade	+11.3	EUR/USD
GBP/USD replication	+10.24 / trade	$p = 0.001$
USD/CAD replication	+4.63 / trade	$p = 0.028$
AUD/USD, NZD/USD replication	failed	excluded from deployment
Sealed 2026-H1 look (out-of-sample)	+7.75 / trade	$n = 16$, $p = 0.29$

The sealed-period p-value of 0.29 is not significant, and we do not claim it is. With 16 trades we cannot distinguish a continuation of the backtest expectancy from a wash. Two more quarters of trading should move the standard error from roughly ±18 pips/trade to ±9.

5.2 Early live trading

The demo account has been live since 2026-06-10. First confirmed automated fill: EUR/USD long, take-profit hit, +$28.10 on a 0.05-lot trade. Order rejections (broker retcode 10027 "algo trading disabled") are logged but do not alter the agent's internal state.

This is too few trades to warrant a results table. The live block in this section will be filled out at the 50-trade and 100-trade marks, both of which are stop-conditions defined in the project roadmap.

6 Limitations

What this report does not claim, and why:

- **The 1.5 target was never optimised.** It was chosen for simplicity. A future grid would test 1.0, 1.25, 1.5, 2.0, 2.5 on the 50-trade live sample first, then on a fresh sealed slice.
- **The sealed-window sample is small.** 16 trades is not a live result; it is a first look. The headline numbers are the walk-forward and cross-pair numbers above.
- **This is one strategy.** The portfolio is diversified across pairs, not across strategy families. A second uncorrelated edge has not yet survived our validation pipeline.
- **Observation features are observation features.** The extension-target ladder (which records where prices reached beyond the 1.5R take-profit), the near-miss vault, and the loss vault all collect data for future studies. None of them feed back into live order placement.
- **We only trade demo.** No real money is at stake. The broker spread model is the broker's own, but the demo environment may overstate fills (no liquidity provider rejection).

7 Conclusion

The headline number is +11.3 pips per trade, median, across 7 of 7 walk-forward out-of-sample windows on EUR/USD, replicated with $p \leq 0.03$ on two of three other pairs, on a frozen parameter set the agent has not changed since deployment. The honest framing is that this is a small but robust edge, not a money printer.

Future work, all gated by the same validation pipeline before any live parameter changes:

- Grid the take-profit multiple after the live sample exceeds 50 trades.
- Test a laddered partial-take-profit policy on the extension ladder's historical data; do not enable it live until it passes walk-forward.
- Add a USD-exposure manager that gates correlated entries when the dollar index moves sharply.
- Promote the EUR/USD daily cell to deployment once it clears the same chain on the sealed slice.

References

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2. Bailey, D. H., Borwein, J. M., López de Prado, M., and Zhu, Q. J. (2014). Pseudo-mathematics and financial charlatanism: the effects of backtest overfitting on out-of-sample performance. *Notices of the American Mathematical Society*, 61(5), 458–471.

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5. Sweeney, R. J. (1986). Beating the foreign exchange market. *Journal of Finance*, 41(1), 163–182.

6. Inner Circle Trader (ICT). Community-taught price-action glossary, accessed via public archives, 2025–2026.

7. Project documentation: `docs/00-journey.md`, `docs/CHECKPOINT.md`, `docs/runbooks/vmware-windows.`

8. Evidence artefacts: `docs/reviews/` (immutable snapshots of walk-forward and cross-pair tests).